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SYSTEM-LEVEL MODELING OF MICROFLOWS IN CIRCULAR AND RECTANGULAR CHANNELS

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ABSTRACT

System-level models for unsteady, incompressible, low Reynolds number flow in channels of slightly varying cross sections of different geometries are presented. The models are based on approximate solution of the unsteady Navier-Stokes equation subject to no-slip and first-order slip boundary conditions. The proposed model, relating the volume rate to the average pressure drop across the channel, is cast into an electric circuit model that consists essentially of an infinite number of parallel branches in series with a resistor and a nonlinear component. The resistor and the nonlinear element in series account respectively for viscous dissipation in the direction of the flow and for the convective part of the inertia term. The set of parallel branches captures the unsteady behavior as well as viscous dissipation normal to flow direction. Previous channel models available in the literature proved to be a special case of the models proposed in this paper at steady state or in the limit of vanishing Reynolds number. The proposed models offer superior accuracy when transient behavior and associated dynamic characteristics are of interest. The models also become more accurate for flows in slightly divergent channels, flows at larger Reynolds number, and in flows that experience sudden changes or that are subjected to forced oscillations with large values of the Strouhal number. The proposed models are flexible in the sense that accuracy and cost can be easily traded by increasing or decreasing the number of branches included in the model. The derived models are compared with previous models and with numerical solutions of the full Navier-Stokes equations.

NOMENCLATURE

A	Channel cross sectional area.
h	Height of rectangular channel cross section.
Kn	Knudsen number.
L	Channel length.
p	Pressure.
Q	Volume flow rate.
r	Radial coordinate for circular channel.
R	Radius of circular channel cross section.
Re	Reynolds number.
St	Strouhal number.
t	Time.
u	Velocity component in flow direction
w	Width of rectangular channel cross section.
x	Coordinate along channel width.
y	Coordinate along channel height.
z	Coordinate in flow direction.
$\mathbf{u}$	Velocity vector.
λ	Mean free path.
μ	Dynamic viscosity.
ρ	Density.
σ_v	Accommodation coefficient.

INTRODUCTION

Research in microfluidics has dramatically increased over the past decade showing its great potential significance in a wide

range of technologies such as pressure sensors, medical diagnostic systems and chemical synthesis. Integrated microfluidic systems are the basis for many proposed micro-devices such as the μ TAS [1, 2], PCR devices [3], and digital microfluidics arrays [4–6]. The complexity of these devices makes simulation of the detailed physics an expensive and a time consuming task. Alternative design and analysis approaches based on reduced order modeling [7] are gaining ground due the advantages they offer in terms of speed, cost, and ability to perform analysis and design in an integrated multiple domain “circuit design” environment such as Spice. System level modeling also enables fast sweeping of the design parameters and consequently significantly reducing the number of candidates. Development of efficient yet accurate reduced order models for individual components such as microchannels, micropump, microvalves and micromixers and microheaters is paramount to the viability of system level design approach.

In microfluidics applications, microchannels offer transport of small volumes of the working fluid. Microchannels are used to transport a wide variety of fluid ranging from gases in microengines [8] to biological materials such as DNA samples in biochips such as the PCR chip. As opposed to flow in macroscale channels, liquid flows in microscale channels are always laminar and often approaching the Stokes flow limit, whereas gaseous micro-flows are generally laminar with small but significant value of Reynolds number. Thus for a model to be as general as possible, capturing the unsteady behavior of the flow accurately is essential particularly in applications in which precise control of the flow conditions is required. Most reduced-order models available in the literature account for steady incompressible flow through channels with no inertia effects, with the corresponding circuit model presented in Fig. 1. The model of Fig. 1 relates the volume flow rate to the pressure drop via the hydraulic resistance which depends on the channel shape and cross-sectional area. Tab. 1 lists expressions for the hydraulic resistance in terms of the channels geometrical parameters and flow properties for channels of rectangular and circular cross sections. Other mod-

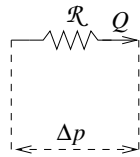


Figure 1. ELECTRIC MODEL OF A STEADY INERTIA-FREE CHANNEL FLOW.

els account for inertia by basing the model on conservation of energy [9, 10] over a control volume $\mathcal{V}$ that consists of the fluid

in the channel bounded by control surface $\mathcal{S}$,

$$\frac{\partial}{\partial t} \int_{\mathcal{V}} \left(\hat{u} + \frac{|\mathbf{u}|^2}{2} + gz \right) \rho d\mathcal{V} + \int_{\mathcal{S}} \left(\hat{h} + \frac{|\mathbf{u}|^2}{2} + gz \right) \rho (\mathbf{u} \cdot \hat{\mathbf{n}}) dS = \dot{Q} - \dot{W}_s - \dot{W}_v$$

where $\hat{u}$ is the specific internal energy, $\mathbf{u}$ is the velocity, g is gravity, z is elevation, $\hat{h}$ is specific enthalpy, $\hat{\mathbf{n}}$ is the outward unit vector normal to $\mathcal{S}$, and $\dot{Q}$, $\dot{W}_s$, $\dot{W}_v$ are respectively the heat gain, the work done by viscous forces, and the work done by body forces. If the flow is isothermal and incompressible, the total internal energy in the pipe is constant and the net flux of internal energy across the control surface is zero, even if the flowrate is time dependent. Assuming also that the pressure, kinetic energy and potential energy per unit mass per unit time are uniform over inlet and outlet, the energy equation reduces to

$$\frac{\partial}{\partial t} \int_{\mathcal{V}} \frac{1}{2} |\mathbf{u}|^2 \rho d\mathcal{V} + \rho Q \left(\frac{p}{\rho} + \frac{1}{2} |\mathbf{u}|^2 + gz \right)_2 - \rho Q \left(\frac{p}{\rho} + \frac{1}{2} |\mathbf{u}|^2 + gz \right)_1 = -\dot{W}_v$$

where Q is the volume flow rate. Dividing by $\rho Q g$, and setting $|\mathbf{u}| = Q/A(z)$ and $d\mathcal{V} = A(z) dz$ where $A(z)$ is the cross section normal to flow coordinate z , the pressure drop $\Delta p \equiv p_1 - p_2$ (across variable) can be related to the volume flow rate Q (through variable) by

$$\Delta p = \rho \frac{dQ}{dt} \int_0^L \frac{dz}{A} + \frac{\rho Q^2}{2} \left(\frac{1}{A_2^2} - \frac{1}{A_1^2} \right) + \rho g(z_2 - z_1) + \frac{\dot{W}_v}{Q} \quad (1)$$

Channel type and geometry	$\mathcal{R}$
Rectangular $w/h \rightarrow \infty$, $h = \text{const}$	$\frac{12\mu L}{wh^3}$
Converging/Diverging 2D rectangular	$\frac{6\mu L(h_1+h_2)}{w(h_1h_2)^2}$
Circular of constant cross section	$\frac{8\mu L}{\pi R^4}$

Table 1. HYDRAULIC RESISTANCE FOR DIFFERENT CHANNEL GEOMETRIES

Eqn. (1) implies that a system level model relating the pressure drop to the volume flow rate consists of an inductor of inductance $(\rho \int_0^L dz/A(z))$ in series with a nonlinear component representing inertia, in series with a component, $\dot{W}_v/Q$,

representing flow resistance due to viscous effects. The work done per unit volume by viscous effects still needs to be determined from the velocity distribution, which requires solving the momentum equation in differential form. The most commonly used approximation found in standard textbooks [11] is based on assuming that the work done per unit volume by viscous effects is obtained under steady inertia free conditions, $\dot{W}_v/Q = \mathcal{R}_{\text{inertia free}}Q(t)$, where $\mathcal{R}$ is the resistance found in Tab. 1. Under this approximation, Eqn. (1) becomes

$$\Delta p = \left(\rho \int_0^L \frac{dz}{A(z)} \right) \frac{dQ}{dt} + \frac{1}{2} \left(\frac{1}{A_2^2} - \frac{1}{A_1^2} \right) \rho Q^2 + \rho g(z_2 - z_1) + R_{\text{inertia free}} Q \quad (2)$$

The model of Eqn. (2) corresponds to the circuit shown in Fig. 2 to a resistance $\mathcal{R} = \mathcal{R}_{\text{inertia free}}$, accounting for viscous effects, in series with an inductance $\mathcal{L} = \rho \int_0^L \frac{dz}{A(z)}$, accounting for the time variation component of the inertia term, in series with a nonlinear term accounting for the convective component of the inertia term. Note that Eqn. (1) may also be obtained by integrating the momentum equation in differential form along a streamline extending from cross section 1 to cross section 2.

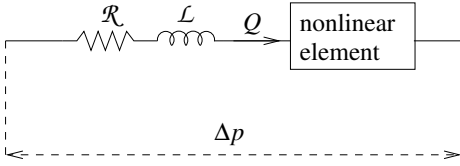


Figure 2. ELECTRIC MODEL OF UNSTEADY FLOW IN A CHANNEL.

For a channel flow due to harmonic pressure drop $\Delta p = \Delta p_0 \cos(\omega t)$, the impedance $Z = \mathcal{R} + i\omega\mathcal{L}$, where R is resistance and $\mathcal{L}$ is inductance, for a rectangular channel of constant cross section was derived by Morris et al. [12], based on earlier work by Drake [13], as

$$Z = \frac{\mu L \pi^2}{2h^4} \sum_{n=0}^{\infty} \frac{1}{(2n+1)^2 q_n^2} \left(\alpha - \frac{\tanh q_n \alpha}{q_n} \right)$$

where $\alpha = w/h$ and $q_n = \sqrt{i \frac{h^2 \omega}{4\nu} + \frac{(2n+1)^2 \pi^2}{4}}$. Fan et. al [14] presented exact solutions for unsteady laminar incompressible flow through rectangular channels of constant cross section. Since the cross section is constant, the inertia term is zero and the solution of [14] is a special case of the solution presented in this paper for channel cross section that varies in the flow direction.

The objective of this work is to assess the accuracy of the model given by Eqn. (2) and to derive more accurate system-level models for unsteady laminar flows in nearly parallel channels of rectangular and circular cross sections. The derived models are compared to numerical results from CFD simulations using Fluent.

GOVERNING EQUATIONS AND BOUNDARY CONDITIONS

The variation of the cross section $A(z)$ in the flow direction z is assumed small in the sense that the velocity components normal to the flow are small compared to the velocity component u in the flow direction. The component of the Navier-Stokes equation in the flow direction is then given by [9, 10]

$$\rho \left(\frac{\partial u}{\partial t} + \mathbf{u} \cdot \nabla u \right) = -\frac{\partial p}{\partial z} + \mu \nabla^2 u \quad (3)$$

where, without loss of generality in the derivations that follow, body forces including gravity are neglected. The momentum equation (3) is complemented with the equation governing conservation of mass, which for constant density ρ , is given by

$$\nabla \cdot \mathbf{u} = 0 \quad (4)$$

Alternatively, conservation of mass may be expressed as

$$\frac{\partial Q}{\partial z} = 0, \text{ where } Q(t) = \int_{A(z)} u dA \quad (5)$$

stating that the volume flow rate Q is a function of time only. It may be shown by order of magnitude analysis that the diffusion component in the flow direction may be neglected when compared to that normal to the flow, i.e. $|\partial^2 u / \partial z^2| / |\nabla^2 u - \partial^2 u / \partial z^2| \sim (A/L^2)$. If A/L^2 is not small enough, then a first order approximation of the axial diffusion terms may be obtained by replacing u with $Q(t)/A(z)$ so that $\partial^2 u / \partial z^2 \simeq Q d^2 A^{-1} / dz^2$. The convection component of the inertia term is approximated by assuming that the velocity component in the flow direction is approximated as $u \simeq (Q(t)/A(z))\phi$ so that

$$\mathbf{u} \cdot \nabla u \simeq \frac{1}{2} \frac{d}{dz} \left(\frac{Q^2(t) \phi^2}{A^2(z)} \right)$$

where ϕ is a function of the coordinates in the plane of the channel cross section and must satisfy the condition $\int_{A(z)} \phi dA = A(z)$. Two particular expressions for ϕ are investigated: $\phi = 1$ which

assumes that in computing the convective term the flow is a plug flow and $\phi = \phi^*$ which assumes in computing the convective term that the flow assumes the parabolic profile ($Q\phi^*/A$) of the fully developed flow. The momentum Eqn. (3) may then be approximated as

$$\frac{1}{v} \frac{\partial u}{\partial t} + \frac{1}{v} f = -\frac{1}{\mu} \frac{\partial p}{\partial z} + \nabla_{\perp}^2 u, \quad (6)$$

where $\nabla_{\perp}^2 u \equiv \nabla^2 u - \partial^2 u / \partial z^2$ and $f = \frac{Q^2(t)}{2} \frac{d}{dz} \left(\frac{\phi^2}{A^2(z)} \right) - vQ(t) \frac{d^2}{dz^2} \left(\frac{1}{A(z)} \right)$.

In this paper, reduced models for unsteady, low Re , compressible flow in channels with slightly varying cross sections for the configurations displayed in Fig. 3 are presented. For each cross section, approximate models for no slip as well as first order slip velocity boundary conditions are presented:

no-slip: $u = 0$ at solid boundary

first order slip: $u - \frac{2 - \sigma_v}{\sigma_v} \lambda \left(\frac{\partial u}{\partial n} \right) = 0$ at solid boundary

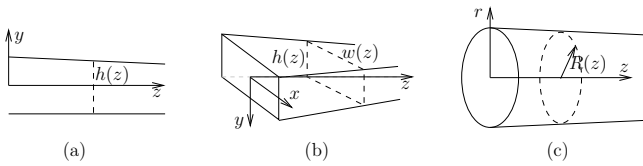


Figure 3. CHANNEL GEOMETRIES CONSIDERED: (a) 2D CHANNEL OF INFINITE WIDTH, (b) CHANNEL OF RECTANGULAR CROSS SECTION, AND (c) CHANNEL OF CIRCULAR CROSS SECTION.

THE MODELS

In this section, system level models of unsteady incompressible flow through a channel of variable cross section are presented for the various configurations of Fig. 3. The models are presented in terms of ordinary differential equation(s) relating the volume flow rate (through variable) to the pressure drop (across variable).

Rectangular channel of infinite width

Derivation of the model for flow in the 2D channel of infinite width depicted in Fig. 3-(a) is presented in Appendix A. The

volume flow rate per unit width is given by

$$Q = \sum_{k=0}^{\infty} Q_k$$

and the ordinary differential equation relating Q , Q_k and Δp for the cases of no-slip and first-order slip boundary conditions are:

No-slip boundary condition

$$\begin{aligned} & \frac{\rho \alpha_k^2}{8} \left(\int_0^L \frac{dz}{h} \right) \frac{dQ_k}{dt} + \frac{\mu \alpha_k^4}{8} \left(\int_0^L \frac{dz}{h^3} \right) Q_k - \mu \left(\frac{dh^{-1}}{dz} \right)_{z_1}^{z_2} Q \\ & + \frac{\rho}{2} \left(\frac{g(\phi^2)}{h^2} \right)_{z_1}^{z_2} Q^2 = \Delta p \end{aligned} \quad (7)$$

where $\alpha_k = (2k+1)\pi$ and $g(\cdot)$ is defined in Eqn. (16).

First-order slip boundary condition

$$\begin{aligned} & \frac{\rho}{2} \left(\int_0^L \frac{\gamma_k dz}{h} \right) \frac{dQ_k}{dt} + \frac{\mu}{2} \left(\int_0^L \frac{\beta_k'^2 \gamma_k dz}{h^3} \right) Q_k - \mu \left(\frac{dh^{-1}}{dz} \right)_{z_1}^{z_2} Q \\ & + \frac{\rho}{2} \left(\frac{g(\phi^2)}{h^2} \right)_{z_1}^{z_2} Q^2 = \Delta p \end{aligned} \quad (8)$$

where β_k' are positive roots of,

$$\tan \beta' = \frac{2Kn' \beta'}{Kn'^2 \beta'^2 - 1},$$

$Kn' = (Hh)^{-1}$, $H = \frac{\sigma_v}{\lambda(2-\sigma_v)}$, and

$$\gamma_k = \frac{2Kn' + (Kn'^2 \beta_k'^2 + 1)}{\left(Kn' \sin \beta_k' + \frac{(1 - \cos \beta_k')}{\beta_k'} \right)^2}$$

Note both β_k' and γ_k are function of z , and that the integrals in Eqns. (7) and (8) are a function of geometry and need only be computed once.

The unsteady flow rate-pressure drop behavior governed by Eqns. (7) and (8) may be represented by a circuit model. According to Eqns. (7) and (8), the flow rate Q may be taken as the through variable (current) and the pressure difference Δp as the across variable (potential difference), and the dynamic

model of the channel flow may then be represented as a set of infinite branches in parallel, in series with a nonlinear component and a resistor, as seen in Fig. 4. For the no-slip model, branch k consists of an inductor of inductance $\mathcal{L}_k = \frac{\rho\alpha_k^2}{8} \left(\int_0^L \frac{dz}{h} \right)$ and a resistance $\mathcal{R}_k = \frac{\mu\alpha_k^4}{8} \left(\int_0^L \frac{dz}{h^3} \right)$. The subcircuit consisting of the parallel branches, is in series with a resistance $\mathcal{R}_0 = \mu \left[(dh^{-1}/dz)_1 - (dh^{-1}/dz)_2 \right]$ and a nonlinear component that is characterized by a pressure drop proportional to the flow rate squared. The potential difference across each branch $k = 0 \dots \infty$ is $\Delta p - \mathcal{R}_0 Q - \kappa Q^2$, where $\kappa = (\rho/2)[g(\phi_2^2)/h_2^2 - g(\phi_1^2)/h_1^2]$, and the current passing through branch k is Q_k . The total current (flow rate) is $Q = \sum_{k=0}^{\infty} Q_k$.

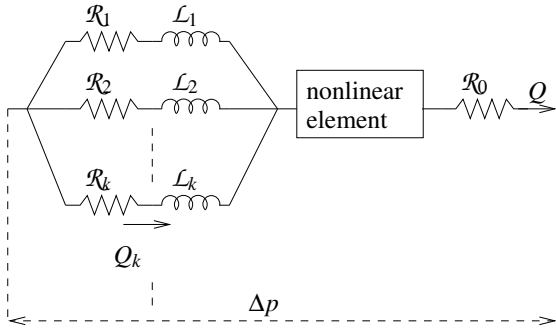


Figure 4. PROPOSED MODEL OF UNSTEADY CHANNEL FLOW.

Dividing Eqn. (7) by α_k^4 and summing for $k = 0 \dots \infty$, then dividing by $C \equiv \sum_{k=0}^{\infty} \alpha_k^{-4} = 1/96$, the no-slip steady behavior is obtained by setting the $d()/dt = 0$ so that

$$\frac{\rho}{2} \left(\frac{g(\phi^2)}{h^2} \right)_{z_1}^{z_2} Q_s^2 + \mu \left[12 \int_0^L \frac{dz}{h^3} - \left(\frac{dh^{-1}}{dz} \right)_{z_1}^{z_2} \right] Q_s = \Delta p \quad (9)$$

Eqn. (9) reduces to the well known correlation between the volume flow rate per unit width and pressure drop for fully developed steady flow in a rectangular channel of constant height h , $Q_s = \Delta p h^3 / (12\mu L)$ which corresponds to a resistance $\mathcal{R} = 12\mu L / h^3$.

At steady state, the functions ϕ_1 and ϕ_2 are obtained from assumed velocity distributions at channel inlet and outlet respectively. If the velocity is uniform such as the idealized case of smooth inlet to channel from a reservoir, then $\phi_1 \simeq 1$ at which $g(\phi_1) = 1$. If, on the other hand, the flow is fully developed because the channel is preceded by another channel, the parabolic velocity profile $\phi = 6(y/h)(1 - y/h)$ yields $g(\phi) \simeq g_1(\phi) = 12/\pi^2$. At outlet, it is assumed that the flow is fully developed so that $g(\phi_2) = 12/\pi^2$. Note that except for

the coefficient g , both Eqns. (9) and (2) provide the same relation between the volume flow rate and the pressure difference at steady state.

Channel of circular cross section

For channels of circular cross section, the ordinary differential equation relating $Q = \sum_{m=1}^{\infty} Q_m$, Q_m and Δp for the cases of no-slip and first-order slip boundary conditions are:

No-slip boundary condition

$$\frac{\rho\gamma_m^2}{4} \left(\int_0^L \frac{dz}{A} \right) \frac{dQ_m}{dt} + \frac{\pi\mu\gamma_m^4}{4} \left(\int_0^L \frac{dz}{A^2} \right) Q_m - \mu \left(\frac{dA^{-1}}{dz} \right)_{z_1}^{z_2} Q + \frac{\rho}{2} \left(\frac{g(\phi^2)}{A^2} \right)_{z_1}^{z_2} Q^2 = \Delta p \quad (10)$$

where $A = \pi R(z)^2$ is the cross sectional area, γ_m are the positive roots of $J_0(\gamma_m) = 0$, J_0 is the modified Bessel function of order 0. Note that for steady flow, the flow rate is obtained by solving the quadratic equation

$$\frac{1}{2} \left(\frac{g(\phi^2)}{A^2} \right)_{z_1}^{z_2} \left(\sum_{m=1}^{\infty} \frac{1}{\gamma_m^4} \right) Q^2 + \mu \left[\frac{\pi}{4} \int_0^L \frac{dz}{A^2(z)} - \frac{1}{32} \left(\frac{dA^{-1}}{dz} \right)_{z_1}^{z_2} \right] Q - \frac{\Delta p}{32} = 0,$$

which, for constant radius R , reduces to the well-known solution $Q_s = (\Delta p / L) (\pi R^4) / (8\mu)$. The circuit model representation of a circular channel (Fig. 4) is similar to that of the rectangular channel of infinite width, where branch m is represented as an inductor (time varying component of inertia) in series with a resistor (viscous dissipation). In series with these parallel branches is a resistor (viscous dissipation in flow direction) in series with a nonlinear component (convective component of inertia).

First-order slip boundary condition

$$\frac{\rho}{4} \left(\int_0^L \frac{\gamma_m}{A} dz \right) \frac{dQ_m}{dt} + \frac{\pi\mu}{4} \left(\int_0^L \frac{\beta_m'^2 \gamma_m}{A^2} dz \right) Q_m - \mu \left(\frac{dA^{-1}}{dz} \right)_{z_1}^{z_2} Q + \frac{\rho}{2} \left(\frac{g(\phi^2)}{A^2} \right)_{z_1}^{z_2} Q^2 = \Delta p \quad (11)$$

where β_m' are the positive roots of $\text{Kn}' \beta' J_1(\beta') = J_0(\beta')$, $\gamma_m = (1 + \text{Kn}'^2 \beta_m'^2) \beta_m'$, $\text{Kn}' = (RH)^{-1}$, and $H = \frac{\sigma_v}{\lambda(2-\sigma_v)}$. Note that β_m' and γ_m are both functions of z and the integrals need to be computed once since they are a function of only the geometry.

Channel of rectangular cross section

If the channel cross section width to height ratio is finite, then diffusion effects along both the height and width must be included. In this section, system-level models for a flow in a channel of rectangular cross section are presented. The ordinary differential equations relating the volume flow rate Q_{mn} through branch (m,n) , the total volume flow rate $Q = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} Q_{mn}$, and the pressure drop for the cases of no-slip and first-order slip boundary conditions are:

No-Slip Boundary Condition

$$\rho \frac{\beta_m^2 \alpha_n^2}{64} \left(\int_0^L \frac{dz}{A} \right) \frac{dQ_{mn}}{dt} + \frac{\mu \beta_m^2 \alpha_n^2}{64} \left(\int_0^L \frac{h^2 \beta_m^2 + w^2 \alpha_n^2}{A^3} dz \right) Q_{mn} - \mu \left(\frac{dA^{-1}}{dz} \right)_{z_1}^{z_2} Q + \frac{\rho}{2} \left(\frac{g(\phi^2)}{A^2} \right)_{z_1}^{z_2} Q^2 = \Delta p \quad (12)$$

where $\beta_m = m\pi$, $\alpha_n = n\pi$ and $A = w(z)h(z)$.

First-order boundary condition

$$\frac{\rho}{4} \left(\int_0^L \frac{\gamma_m \eta_n}{A} dz \right) \frac{dQ_{mn}}{dt} + \frac{\mu}{4} \left(\int_0^L \frac{\gamma_m \eta_n (w^2 \beta_m'^2 + h^2 \alpha_n'^2)}{A^3} dz \right) Q_{mn} - \mu \left(\frac{dA^{-1}}{dz} \right)_{z_1}^{z_2} Q + \frac{\rho}{2} \left(\frac{g(\phi^2)}{A^2} \right)_{z_1}^{z_2} Q^2 = \Delta p \quad (13)$$

where

$$\gamma_m = \frac{2Kn_h + (Kn_h^2 \beta_m'^2 + 1)}{\left(Kn_h \sin \beta_m' + \frac{(1 - \cos \beta_m')}{\beta_m'} \right)^2}$$

$$\eta_n = \frac{2Kn_w + (Kn_w^2 \alpha_n'^2 + 1)}{\left(Kn_w \sin \alpha_n' + \frac{(1 - \cos \alpha_n')}{\alpha_n'} \right)^2},$$

β_m' and α_n' are respectively the positive roots of

$$\tan \beta' = \frac{2Kn_h \beta'}{Kn_h^2 \beta'^2 - 1}, \quad \tan \alpha' = \frac{2Kn_w \alpha'}{Kn_w^2 \alpha'^2 - 1}$$

and

$$Kn_h = \frac{2 - \sigma_v}{\sigma_v} \frac{\lambda}{h}, \quad Kn_w = \frac{2 - \sigma_v}{\sigma_v} \frac{\lambda}{w}$$

MODEL VALIDATION AND DISCUSSION

Accuracy and performance of the models presented in the previous section are addressed in terms of steady state and transient analyses of various examples. The models enables system level analysis of channel flow using the conventional techniques employed in circuit analysis. Direct Current (DC) analysis is steady state analysis in which the transient term is set to zero and all the inductors in Figure 4 are short circuited. Transient analysis can also be performed by integrating the ODEs in time using circuit simulation tools such as Spice. In all of the analyses that follow, unless otherwise specified, the density, viscosity, and reference length are taken to be unity so that $Re = Q$.

Steady state analyses

Rectangular channel of infinite width Fig. 6 shows steady state results for fully developed flow in a convergent channel of cross section $h = h_1 + z(h_2 - h_1)/L$, with $L = 10$. A schematic is depicted in Fig. 5. The results of Eqn. (9) with $g(0) = g(L) = 12/\pi^2$ compare well with converged numerical results obtained using Fluent for the geometry shown in Fig. 5. Comparison for steady fully developed flow in a divergent chan-

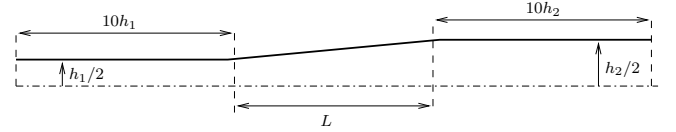


Figure 5. GEOMETRY USED FOR CFD SIMULATIONS IN FLUENT.

nel presented in Fig. 7 show that deviations of the model of Eqn. (9) from the numerical results of Fluent become more prominent for larger $(h_2 - h_1)/L$.

Channel of circular cross section Steady state solutions of Eqn. (10) for flow through converging channel versus static pressure drop are presented in Fig. 8 for the various choices of ϕ and compared with solution obtained by numerical simulations using Fluent. The results indicate that $\phi = 2(1 - r^2/R^2)$ yields the most accurate solution of the model. Similar results for diverging channel plotted in Fig. 9 yield the same conclusion.

Note that the convection (nonlinear) term reduces to zero for the case where inlet and outlet cross sections are equal, $A(z_1) = A(z_2)$, even though the cross sectional area may vary in between. As an example, a circular channel with a cross section that varies in the flow direction according to $R = R_0 + \delta \sin \frac{2\pi fz}{L}$ is considered, with $R_0 = 0.5$, $\delta = 0.1$, $f = 5$, $L = 10$. In this case $A(z = 0) = A(z = L)$ and consequently the convective (nonlinear)

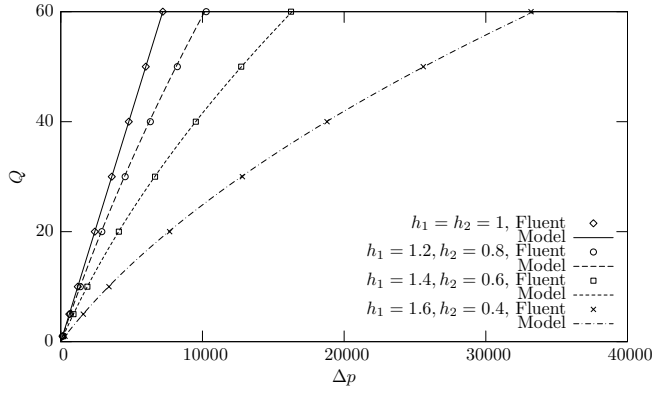


Figure 6. STEADY VOLUME FLOW RATE VERSUS PRESSURE DROP: COMPARISON BETWEEN PROPOSED MODEL AND FLUENT FOR CONVERGING CHANNELS.

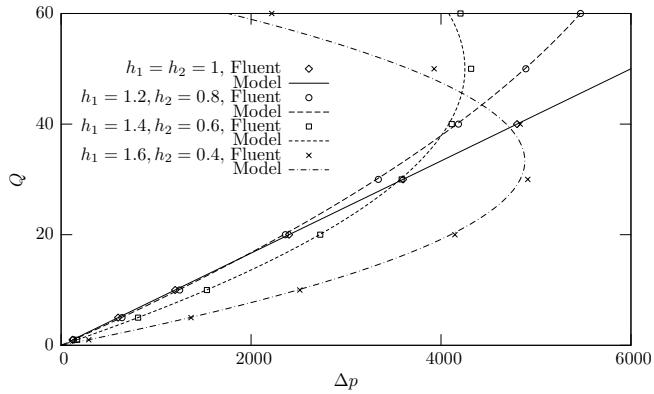


Figure 7. STEADY VOLUME FLOW RATE VERSUS PRESSURE DROP: COMPARISON BETWEEN PROPOSED MODEL AND FLUENT FOR DIVERGING CHANNELS.

term is zero. This introduces significant error in the steady solution as shown in Fig. 10 where the steady solution of Eqn. (10), shown as the dashed line with + symbols, is considerably off of the numerical solution (thick solid line) predicted by Fluent. Presented on the same plot is a fitted solution ($\square$ symbols) obtained by setting the convection term in Eqn. (10) with $\frac{\rho g}{2} c Q^2$, where $c = 3.25$. In order to accurately predict unsteady behavior, steady state characteristic curve must be accurately predicted, and consequently the convective term is set to $\frac{\rho g}{2} c Q^2$ in the unsteady analysis.

Unsteady Analyses

Rectangular channel of infinite width Even though the models given by Eqns. (7) and (2) are identical at steady state (except for $g(\phi^2)$), they model the transient behavior of unsteady

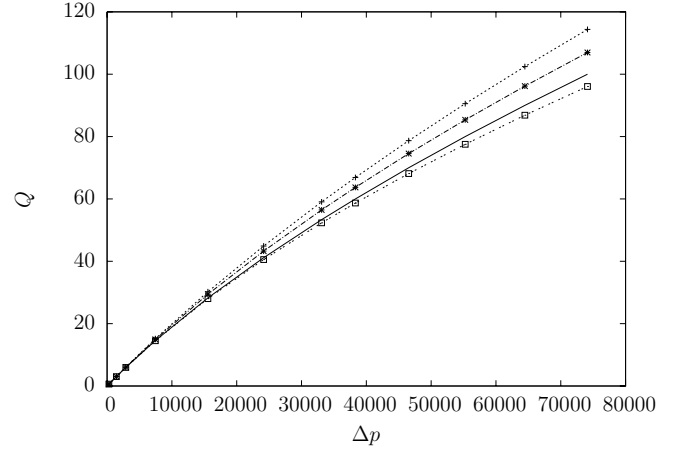


Figure 8. VOLUME FLOW RATE VS. PRESSURE DROP ACROSS A CONVERGING CHANNEL OF CIRCULAR CROSS SECTION OF RADIUS $R = R_1 + (R_2 - R_1)z/L$ WITH $R_1 = 0.6, R_2 = 0.4, L = 10$. SOLID LINE: NUMERICAL (FLUENT). THE + SYMBOLS: $g = 1$, THE * SYMBOL: $g = 8/\gamma_1^2$, AND THE $\square$ SYMBOL: $g = -32\gamma_1^{-2} + 256\gamma_1^{-4}$.

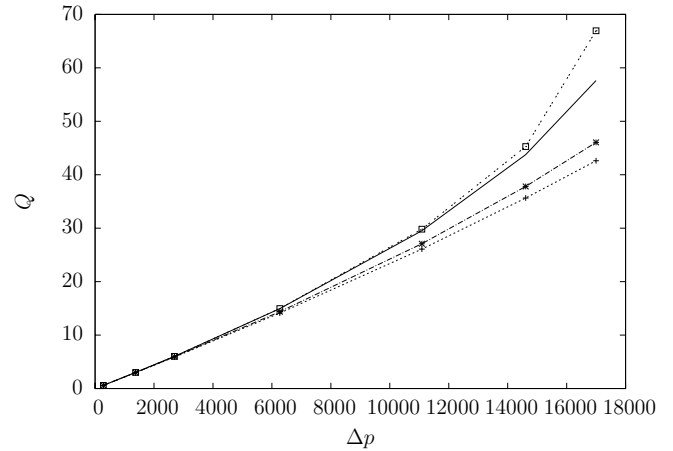


Figure 9. VOLUME FLOW RATE VS. PRESSURE DROP ACROSS A DIVERGING CHANNEL OF CIRCULAR CROSS SECTION OF RADIUS $R = R_1 + (R_2 - R_1)z/L$, WITH $R_1 = 0.4, R_2 = 0.6, L = 10$. SOLID LINE: NUMERICAL (FLUENT). THE + SYMBOLS: $g = 1$, THE * SYMBOL: $g = 8/\gamma_1^2$, AND THE $\square$ SYMBOL: $g = -32\gamma_1^{-2} + 256\gamma_1^{-4}$.

flows differently. While model (2) approximates the time varying term as $\frac{\partial u}{\partial t} \simeq \frac{1}{A(z)} \frac{dQ}{dt}$, the model proposed in this paper allows for a more accurate account by solving the unsteady equation for the velocity and consequently the volume flow rate, as laid out in the previous section. Accuracy of the two models for unsteady flows is investigated by comparing with numerical solution of the Navier-Stokes equations for impulsively started flows in con-

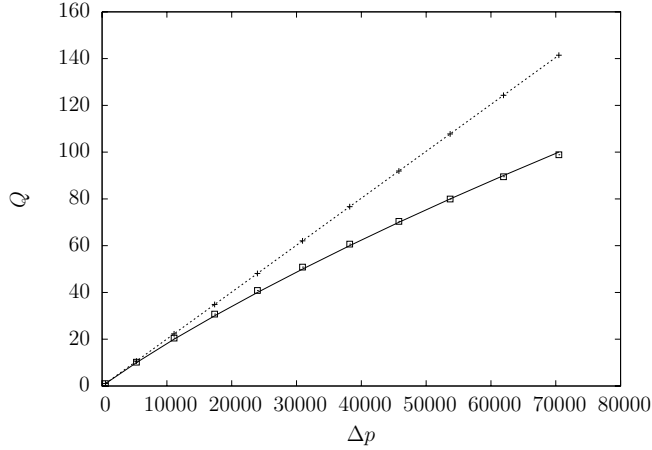


Figure 10. VOLUME FLOW RATE VS. PRESSURE DROP ACROSS A CHANNEL OF CIRCULAR CROSS SECTION OF RADIUS $R = R_0 + \delta \sin \frac{2\pi fz}{L}$ WITH $R_0 = 0.5, \delta = 0.1, f = 5, L = 10$. SOLID LINE: NUMERICAL (FLUENT), THE + SYMBOLS: CONVECTIVE TERM $\frac{\rho g}{2} \left(\int_0^L \frac{dA^{-2}}{dz} dz \right) Q^2 = 0$, AND THE $\square$ SYMBOL: NONLINEAR TERM $\frac{\rho g}{2} c Q^2$.

verging channels (Fig. 11) and in diverging channels (Fig. 12). Fig. 11 shows the time evolution of the volume flow rate (per unit depth) in response to a step function in the difference between stagnation pressure at the inlet to the channel (at the right side for converging channel and at the left side for a diverging channel) of Fig. 5 and outlet static pressure, $p_{0,in} - p_{out} = A \mathcal{H}(t)$, where $A = 50000$ and $\mathcal{H}(\cdot)$ is the Heaviside function. The length of the converging section of the channel is $L = 10$ and inlet and outlet heights of the converging section of the channel are h_1 and h_2 respectively. Similarly, Fig. 12 shows the time evolution of the volume flow rate (per unit depth) in response to a step function in the difference $p_{0,in} - p_{out} = A \mathcal{H}(t)$ across the diverging channel. The Δp time histories for the two cases are obtained from the numerical solution of the Navier-Stokes equations, and are used as the input to the models (Eqns. (7) and (2)). This guarantees that the comparison of the volume flow rate history between the models and the CFD results is based on the same static pressure drop history across the converging or diverging section. The computational geometry is chosen such that the flow is fully developed at the inlet to the converging channel section or to the diverging channel section. Noting that the properties are chosen such that $Re = Q$, the delay in volume flow rate response increases as the Reynolds number increases, which is expected since larger Reynolds number indicates larger inertia.

The results of Figures 11 and 12 show that the transient behavior predicted by model (7) is more accurate than that predicted by model (2) for unsteady flow in both the converging and the diverging channels. Transient parameters such as the rise

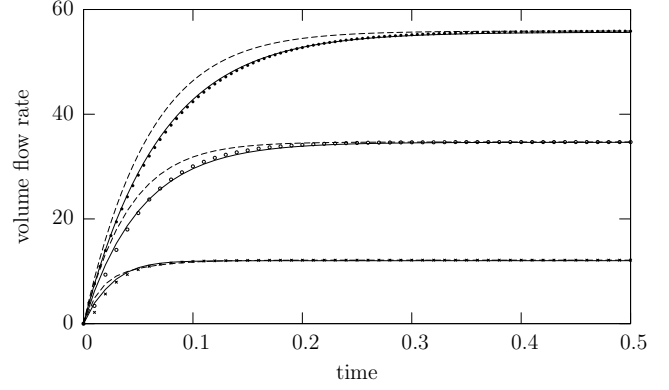


Figure 11. VOLUME FLOW RATE TRANSIENT RESPONSE TO A STEP IN $(p_{0,in} - p_{out})$ ACROSS CONVERGING CHANNEL. SOLID LINES: NUMERICAL (FLUENT). AND DASHED LINES: MODEL OF EQN. (2). SYMBOLS: PROPOSED MODEL OF EQN. (7): FILLED CIRCLES: $h_1 = 1.2, h_2 = 0.8$, EMPTY CIRCLES: $h_1 = 1.4, h_2 = 0.6$ AND $\times$: $h_1 = 1.6, h_2 = 0.4$.

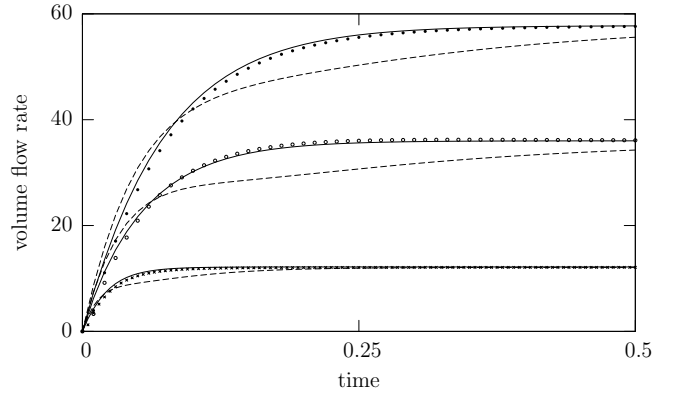


Figure 12. VOLUME FLOW RATE TRANSIENT RESPONSE TO A STEP IN $(p_{0,in} - p_{out})$ ACROSS DIVERGING CHANNEL. SOLID LINES: NUMERICAL (FLUENT). AND DASHED LINES: MODEL OF EQN. (2). SYMBOLS: PROPOSED MODEL OF EQN. (7): FILLED CIRCLES: $h_1 = 0.8, h_2 = 1.2$, EMPTY CIRCLES: $h_1 = 0.6, h_2 = 1.4$ AND $\times$: $h_1 = 0.4, h_2 = 1.6$.

time, fall time and settling time are clearly accurately captured by the proposed model of Eqn. (7), in contrast with the results of equation(2) which underpredicts the rise time for the converging channel and significantly underpredicts the rise time for the diverging channel.

Channel of circular cross section Fig. 13 shows time evolution of the volume flowrate across the channel in re-

sponse to sinusoidal static pressure drop across a channel of circular cross section with $R = R_0 + \delta \sin(2\pi f z/L)$ with $R_0 = 0.5, \delta = 0.1, f = 5, L = 10$. The model of Eqn. (10) compares well with the solution obtained numerically using Fluent. The resulting flow rate amplitude is not large enough to have significant inertia effects and as such no correction for the nonlinear term is implemented. The proposed model, however, predicts accurately both the amplitude as well as the phase of the time response.

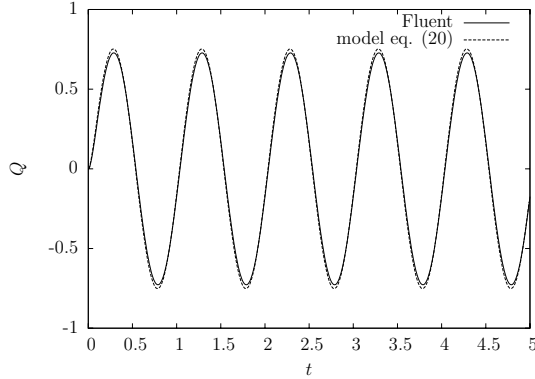


Figure 13. TRANSIENT RESPONSE OF VOLUME FLOW RATE IN RESPONSE TO OSCILLATING PRESSURE DROP.

Channel of rectangular cross section The impact of channel cross-section width to height ratio w/h on the time evolution of the volume flow rate per unit width in response to a pressure step of $\Delta p = 100H(t-1)$ is presented in Figure 14 for $w/h = 1, 2, 4, 20$ and ∞ . The channel length is $L = 10$. The density and viscosity are $\rho = 1$ and $\mu = 1$ so that Reynolds number based on channel height is $Re = Q/w$. As w/h increases, the wetted perimeter per unit width is less resulting in smaller viscous effects so that a larger volume flow rate per unit width is expected. The solution for $w/h \rightarrow \infty$ reduces to that predicted by Eqn. (7) for the rectangular channel of infinite width. The impact of kinematic viscosity on the dynamic behavior of the volume flow rate through a channel of constant cross section in response to a harmonic pressure drop $\Delta p/Re_0 = 1000 \sin(t)$ is shown in Fig. 15 for different values of Reynolds number ($Re_0 = \rho U_0 h/\mu$ with $U_0 = 1$). The channel length is $L = 10$. As can be seen from the figure, larger values of Reynolds number result in larger amplitudes of the volume flow rates. The phase difference between the various responses is due to the fact that flows with larger Reynolds number are characterized by larger inertia effects (relative to viscous effects) so that it takes them more time to adapt to time variations in the pressure drop. Fig. 16 shows the dynamic

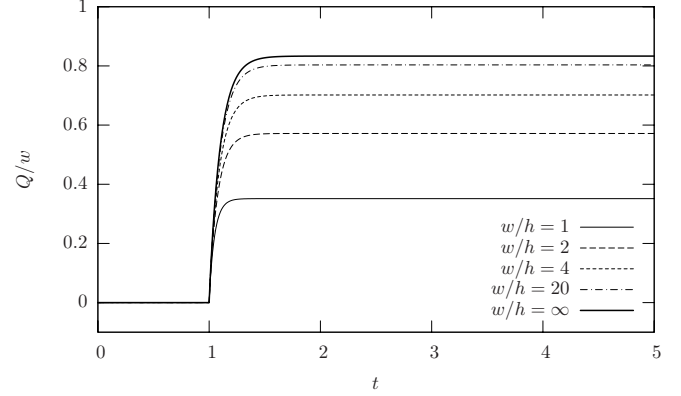


Figure 14. IMPACT OF RATIO OF CHANNEL WIDTH TO HEIGHT ON THE VOLUME FLOW RATE PER UNIT WIDTH. $\rho = 1, \mu = 1, L = 10, \Delta p = 100H(t-1), Kn_0 = 0$.

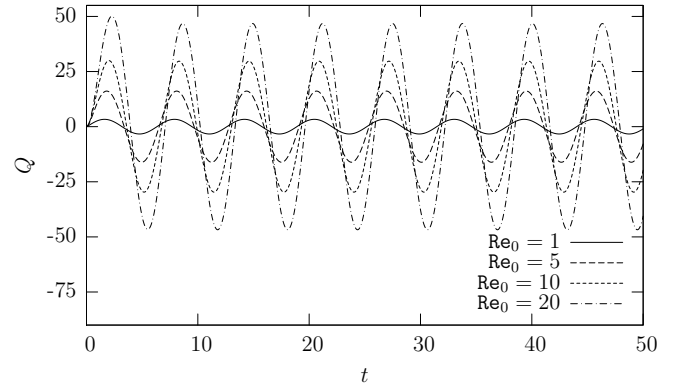


Figure 15. IMPACT OF REYNOLDS NUMBER ON VOLUME FLOW RATE TRANSIENT RESPONSE TO A SINUSOIDAL PRESSURE DROP FOR A CHANNEL OF CONSTANT CROSS SECTION. $Kn_0 = 0, L = 10, \Delta p/Re_0 = 1000 \sin(t), h_1 = w_1 = 1, h_2 = w_2 = 1$.

behavior of the volume flow rate in response to the harmonic pressure difference $\Delta p = 1000 \sin(t)$ for a converging channel, channel with a constant cross section, and diverging channel. All channels of length $L = 10$ have square cross sections and same mean width and mean height ($\bar{h}$). The Reynolds number ($Re_0 = \rho U_0 \bar{h}/\mu$ with $U_0 = 1$) is $Re_0 = 10$. The channel with the largest volume flow rate is that with constant cross section. Note also that flow rate through this channel lags in phase, then the converging channel and then the diverging channel. The difference in phase is due to the fact that inertia effects (relative to viscous losses) in the straight channel are largest and thus the flow rate through the channel takes the longest time to respond to time variations in Δp .

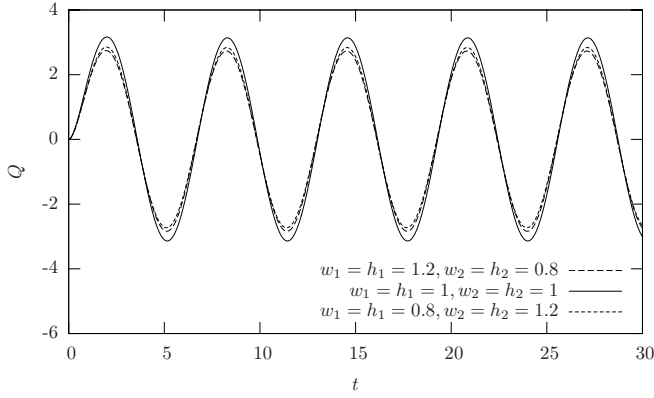


Figure 16. IMPACT OF CHANNEL CROSS SECTION ASPECT RATIO ON VOLUME FLOW RATE TRANSIENT RESPONSE TO A SINUSOIDAL PRESSURE DROP ACROSS A CHANNEL OF RECTANGULAR CROSS SECTION. $Re_0 = 10$, $L = 10$, $\Delta p = 1000 \sin(t)$, $Kn_0 = 0$.

Impact of first-order slip boundary condition

The impact of the Knudsen number ($Kn_0 = Kn'$) on the volume flow rate response to a step function in the pressure difference, $\Delta p = 1000 \mathcal{H}(t - 1)$, is depicted in Fig. 17 for a channel with $h = 1$ and $L = 10$. The steady value of the volume flow rate at a given static pressure drop increases linearly with Knudsen number, which is in accordance with the steady solution

$$\frac{Q_s}{w} = \frac{h^3}{2\mu} \frac{\Delta p}{L} \left(\frac{1}{6} + Kn' \right)$$

The increase in volume flow rate for larger values of Knudsen number implies that inertia effects are playing a bigger role resulting in an increase in the rise time, as observed in Fig. 17. The impact of increasing the Knudsen number on the dynamic behavior of the volume flow rate through a converging channel of a square cross section in response to a harmonic pressure difference $\Delta p = 1000 \sin(t)$ is depicted in Fig. 18. The channel length is $L = 10$, Reynolds number is $Re_0 = \rho U_0 \bar{h} / \mu = 10$ with $U_0 = 1$, inlet cross section height and width are $h_1 = w_1 = 1.2$, and outlet cross section height and width are $h_2 = w_2 = 0.8$. Increasing Knudsen number reduces viscous effects so that larger volume flow rates for the same pressure differential are expected. The dynamic impact of increasing Kn_0 is increasing the phase lag due to the increasing relative effect of inertia, as discussed previously.

CONCLUSION AND FUTURE WORK

Reduced order models for unsteady incompressible flow in nearly parallel channels of circular and rectangular cross sections have been developed. Models for no slip as well as first

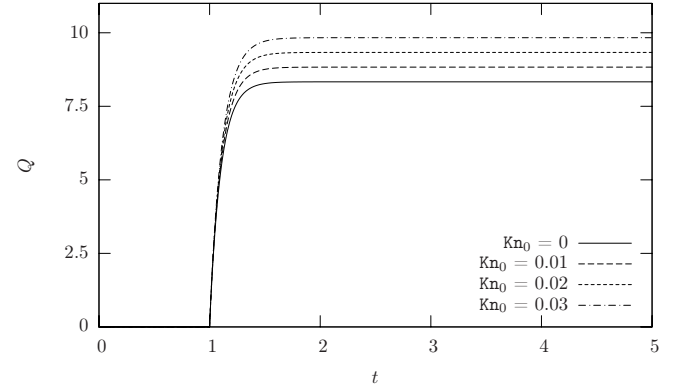


Figure 17. IMPACT OF KNUDSEN NUMBER ON VOLUME FLOW RATE TRANSIENT RESPONSE TO A STEP IN PRESSURE DROP FOR A STRAIGHT CHANNEL. $h = 1$, $\rho = 1$, $\mu = 1$, $L = 10$, $\Delta p = 1000 \mathcal{H}(t - 1)$.

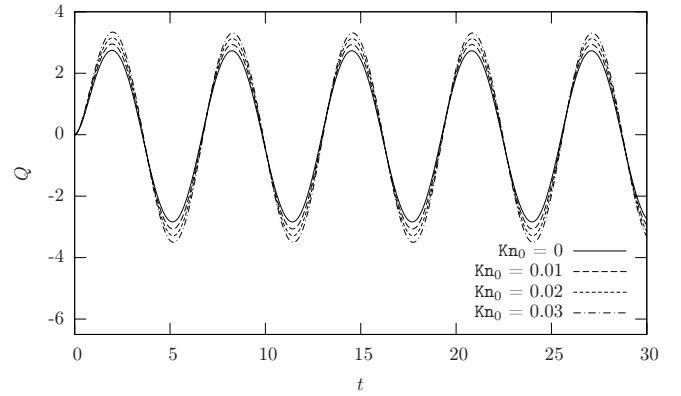


Figure 18. IMPACT OF KNUDSEN NUMBER ON VOLUME FLOW RATE TRANSIENT RESPONSE TO A SINUSOIDAL PRESSURE DROP ACROSS A CONVERGING CHANNEL OF A SQUARE CROSS SECTION. $Re_0 = 10$, $L = 10$, $\Delta p = 1000 \sin(t)$, $h_1 = w_1 = 1.2$, $h_2 = w_2 = 0.8$.

order slip boundary conditions are presented. Each model consists essentially a non-linear component in series with a set of parallel branches, each consisting of a resistance and an inductor in series. While the resistors model viscous dissipation and the inductors correspond to inertia due to the time varying term, the non-linear component accounts for the added inertia due to the convection term. The accuracy and validity of the proposed model are investigated in terms of dynamic response of the volume flow rate for sinusoidal as well as step function pressure history. The proposed models offer more accuracy than currently existing models and a tradeoff between speed and accuracy is possible by controlling the number of parallel branches included

in the model.

The models accuracy for unsteady flow is dependent on the accuracy of the steady solution which is not readily available for general variations of the cross section in the flow direction. Future work will focus on getting more accurate steady solutions for cross sections variations encountered in microfluidic applications.

Acknowledgements

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Appendix A: Derivation of the reduced order for flow through a channel of infinite width with no-slip boundary condition

The z -component of the momentum equation subject to the no-slip velocity at the channel boundaries is

$$\frac{1}{v} \frac{\partial u}{\partial t} + \frac{1}{v} f(z, y, t) = -\frac{1}{\mu} p_z + \frac{\partial^2 u}{\partial y^2} \quad (14)$$

where $p_z = dp/dz$ and $f \simeq \mathbf{u} \cdot \nabla u - v \partial^2 u / \partial z^2$. The solution [15] is that of the heat equation with a source term and is given by

$$u(y, t) = \sum_{k=1}^{\infty} e^{-v\beta_m^2 t} K(\beta_m, y) \int_0^t e^{v\beta_m^2 t'} \chi(\beta_m, t') dt', \text{ where}$$

$$\chi = -\frac{v}{\mu} \int_0^h K(\beta_m, y') (p_z(t') + \rho f(z, y', t')) dy'$$

Integrating from $y = 0..h$, the volume flow rate per unit width is

$$\frac{Q(t)}{w} = \sum_{k=1}^{\infty} e^{-v\beta_m^2 t} F(\beta_m, h) \int_0^t e^{v\beta_m^2 t'} \chi(\beta_m, t') dt'$$

where $F = \int_0^h K dy$. Differentiating with respect to time,

$$\frac{1}{w} \frac{dQ}{dt} = \sum_{k=0}^{\infty} -v\beta_m^2 e^{-v\beta_m^2 t} F(\beta_m, h) \int_0^t e^{v\beta_m^2 t'} \chi(\beta_m, t') dt' + \sum_{k=0}^{\infty} e^{-v\beta_m^2 t} F(\beta_m, h) \frac{d}{dt} \int_0^t e^{v\beta_m^2 t'} \chi(\beta_m, t') dt'$$

and setting $Q_m = w e^{-v\beta_m^2 t} F(\beta_m, h) \int_{t'=0}^t e^{v\beta_m^2 t'} \chi(\beta_m, t') dt'$

where $Q(t) = \sum_m Q_m$, then

$$\frac{dQ}{dt} = \sum_{m=0}^{\infty} \frac{dQ_m(t)}{dt} = \sum_{m=0}^{\infty} -v\beta_m^2 Q_m(t) + w \sum_{m=0}^{\infty} F(\beta_m, h) \chi(\beta_m, t)$$

Noting that $\chi = -\frac{F}{\rho} p_z(t) - \int_0^h K(\beta_m, y') f(z, y', t) dy'$, then

$$\frac{1}{w} \frac{dQ_m(t)}{dt} = -v\beta_m^2 \frac{Q_m(t)}{w} - \frac{F^2(\beta_m, h)}{\rho} p_z(t) - F(\beta_m, h) \int_0^h K(\beta_m, y') f(z, y', t) dy' \quad (15)$$

where the Kernel K is given by $K(\beta_m, y) = \sqrt{\frac{2}{h}} \sin \beta_m y$ and the eigenvalues β_m are the positive roots of $\sin \beta_m h = 0$ leading to $\beta_m = \frac{m\pi}{h}$ for $m = 1, 2, 3, \dots$ so that $F = \frac{2\sqrt{2h}}{(2k+1)\pi}$ for $k = 0, 1, 2, \dots$ Eqn. (15) may be, after integrating from $z = 0$ to $z = L$, rearranged as

$$\frac{\rho}{w} \frac{dQ_m}{dt} \int_0^L \frac{dz}{F^2(\beta_m, h)} + \frac{\mu}{w} Q_m \int_0^L \frac{\beta_m^2}{F^2(\beta_m, h)} dz + \rho \int_0^L F^{-1}(\beta_m, h) g(f) \int_0^h K(\beta_m, y') f(z, y', t) dy' dz = \Delta p$$

where

$$g_m(f) \equiv \frac{\int_0^h K(\beta_m, y') f(z, y', t) dy'}{\int_0^h K(\beta_m, y') dy'} \quad (16)$$

and substituting expressions for F, β_m , and $f \simeq \frac{Q(t)^2}{2} \frac{d}{dz} \left(\frac{\phi^2}{A^2(z)} \right) - \nu Q \frac{d^2}{dz^2} \left(\frac{1}{A(z)} \right)$

$$\frac{\rho \alpha_k^2}{8w} \left(\int_0^L \frac{dz}{h} \right) \frac{dQ_k}{dt} + \frac{\mu \alpha_k^4}{8w} \left(\int_0^L \frac{dz}{h^3} \right) Q_k - \mu \left(\frac{d}{dz} \frac{1}{A(z)} \right)_{z_1}^{z_2} Q + \frac{\rho}{2} \left(\frac{g(\phi_2^2)}{A_2^2} - \frac{g(\phi_1^2)}{A_1^2} \right) Q^2 = \Delta p$$

where $\alpha_k \equiv (2k+1)\pi$, $k = 0, 1, 2, \dots$